

COIM — Collective Outcome Integrity Module

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Independent Methodological Authority

OriginID: OOF-OID-AI-COIM-2026-06-03-0001

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Cognitive Governance Intelligence Architecture (CLIA®)

Operational Layer: Multi-Agent Cognition Governance Layer

Governed Space: Collective Outcome Integrity

Category: AI & Interpretation

Subcategory: Collective Outcome Governance

Type: Multi-Agent Cognition Integrity Module

Parent Standard: Multi-Agent Cognition Integrity Standard (MCIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 3 June 2026

Compatibility: OOF Methodology OS ·

Multi-Agent Cognition Integrity Standard (MCIS) ·

Agent Cognition Integrity Standard (ACIS) ·

Cognitive Synchronization Integrity Module (CSIM) ·

Cognitive Coordination Integrity Module (CCIM) ·

Delegated Cognition Integrity Module (DCIM) ·

Cognitive Conflict Integrity Module (CCFIM) ·

Operational Decision Integrity Standard (ODIS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Collective Outcome Integrity Module (COIM) defines the structural conditions under which outcomes produced by distributed cognition remain collectively attributable, operationally aligned, governance-valid, traceable, consequence-aware, and materially connected to the cognition processes that generated them.

COIM governs collective outcomes.

The module ensures that outputs emerging from multiple agents remain accountable, explainable, and operationally legitimate even when no single agent controls the entire cognition chain.

A system satisfies COIM only if:

- collective outcomes remain attributable
- outcome-generation pathways remain traceable

- collective objectives remain aligned
- consequence ownership remains identifiable
- outcome legitimacy remains reviewable
- collective cognition remains connected to resulting actions

A multi-agent cognition environment that cannot explain how collective outcomes emerged does not satisfy COIM.

Module Operational Role

COIM defines the collective-outcome governance layer of MCIS.

Its role is to preserve accountability and legitimacy for outcomes produced through distributed cognition.

Module Operational Space

COIM governs:

- collective outcomes
- outcome attribution
- consequence ownership
- distributed decision outcomes
- collective cognition accountability
- outcome traceability
- governance-valid outcomes
- collective operational legitimacy

The module applies wherever multiple agents jointly contribute to operational decisions, recommendations, plans, or actions.

Module Function

The module applies wherever systems must preserve:

- accountable outcomes
- traceable result generation
- consequence visibility
- collective legitimacy
- governance-valid execution
- distributed cognition accountability

Its function is to ensure that collective cognition remains connected to collective responsibility.

Minimum Implementation Framework

1. Define the Collective Outcome Object

The organization must define which outcomes require collective-outcome governance.

This may include:

- strategic decisions
 - operational plans
 - workflow outcomes
 - optimization recommendations
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- execution approvals
- autonomous actions
- coordinated responses
- distributed decision outputs

2. Define Collective Outcome Conditions

The system must define the conditions under which collective outcomes remain valid.

This includes:

- attribution requirements
- traceability requirements
- consequence-awareness requirements
- objective-alignment requirements
- accountability requirements
- governance-valid outcome requirements

3. Define Outcome Integrity Degradation Detection Logic

The system must define how collective-outcome integrity degradation is identified.

This may include:

- unattributable outcomes
- broken cognition chains
- objective misalignment
- accountability gaps
- consequence-ownership ambiguity
- governance visibility failures

4. Define Operational Response or Governance Logic

The system must define governance logic for collective-outcome integrity failures.

Governance response may include:

- outcome review
- attribution reconstruction
- accountability reassessment
- governance intervention
- escalation
- execution restriction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- contributing agents
- cognition pathways
- delegation chains
- coordination activities
- governance reviews
- resulting outcomes

A multi-agent cognition environment must not remain outcome-valid if materially significant outcomes cannot be traced back to the cognition processes that generated them.

Use Case 1 — Enterprise Multi-Agent Decision System

Scenario

Multiple specialized agents collaborate to generate recommendations for resource allocation, strategic planning, and operational optimization.

Application

COIM governs attribution, accountability, and traceability of the final collective outcome.

Result

The organization gains stronger governance legitimacy, improved accountability, and reduced risk of unattributable distributed decisions.

Use Case 2 — Autonomous Infrastructure Coordination Network

Scenario

A distributed network of autonomous agents collectively manages operational decisions affecting infrastructure stability, logistics, and service continuity.

Application

COIM governs how collective outcomes are generated, attributed, reviewed, and connected to operational consequences.

Result

The environment gains stronger consequence awareness, improved governance oversight, and reduced exposure to accountability gaps.

Canonical Closing Statement

Collective Outcome Integrity Module (COIM) defines the structural conditions under which outcomes produced by distributed cognition remain collectively attributable, operationally aligned, governance-valid, traceable, consequence-aware, and materially connected to the cognition processes that generated them.

Collective cognition cannot remain valid if collective outcomes lose accountability. Collective outcome integrity therefore becomes a foundational integrity condition of governable multi-agent cognition.

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Canonical Source

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